

a4a stochastic slice biomass index

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1 Intro

[...]

2 Materials and Methods

[...]

2.1 Data

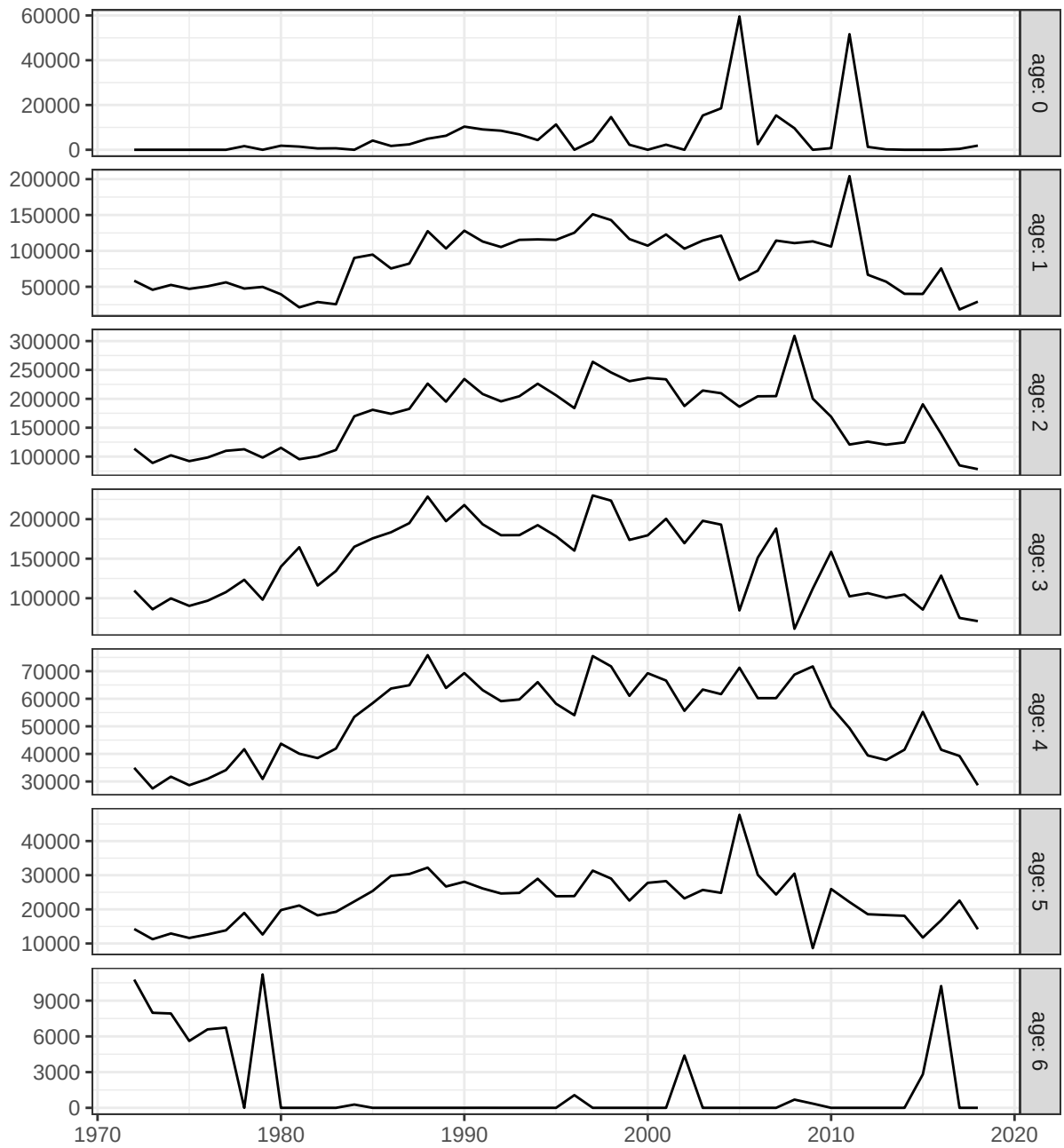


Figure 1. Time series of catch-at-age.

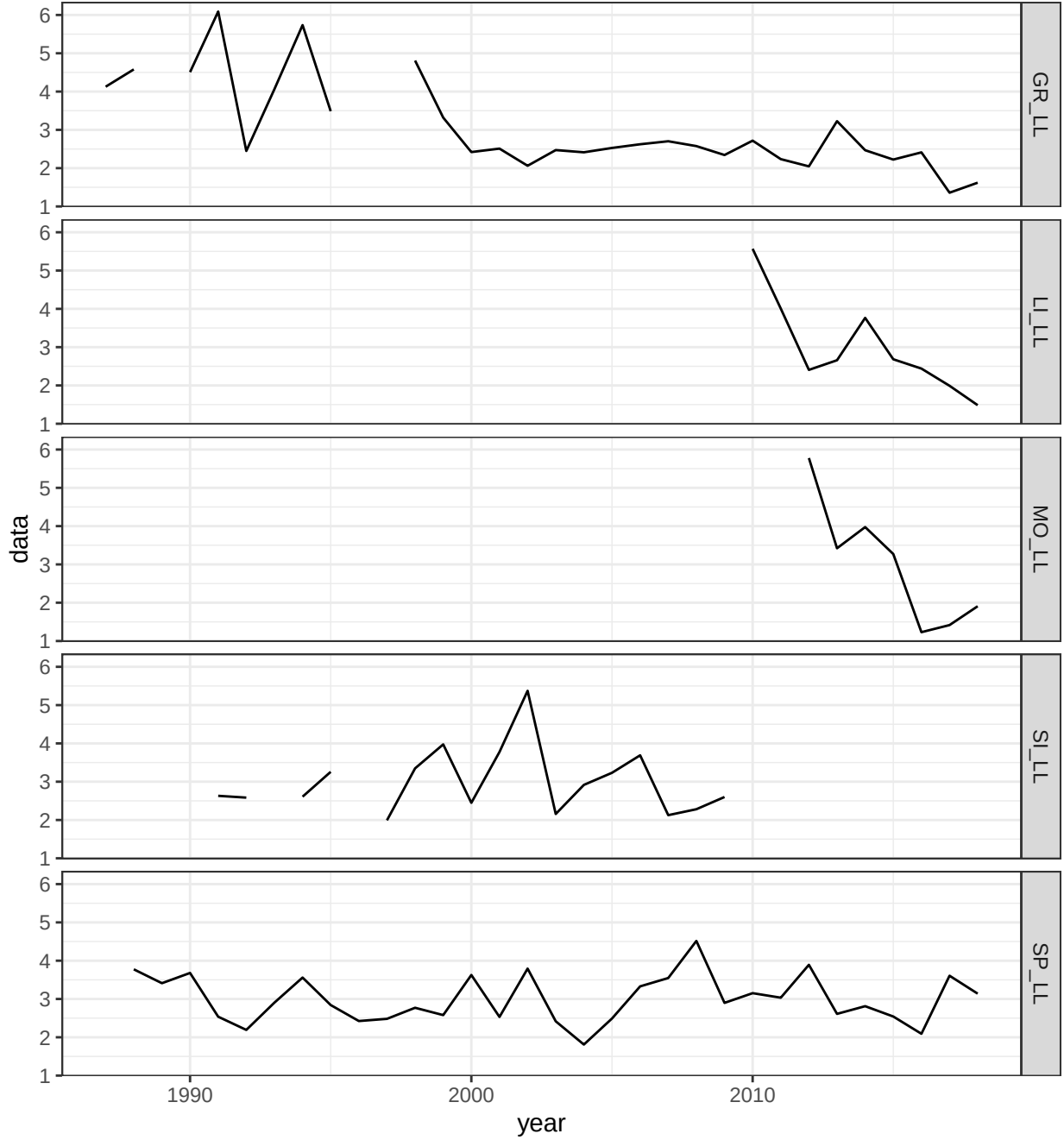


Figure 2. Biomass indices.

3 Results

3.1 Full time series

The simple separable model, treating years as categorical variables gave unreasonable results in both recruitment and f rates.

Without restraining the fishing mortality model to be the same after age 4 we get a very low f and a very high ssb , variants of the separable splines modeling on year gives pretty much the same results. (varying k

around the half of the range of time series). On the other hand the varying degrees of freedom in smoothing splines on age, gives a completely different perception of the stock, which suggests a quite unstable model.

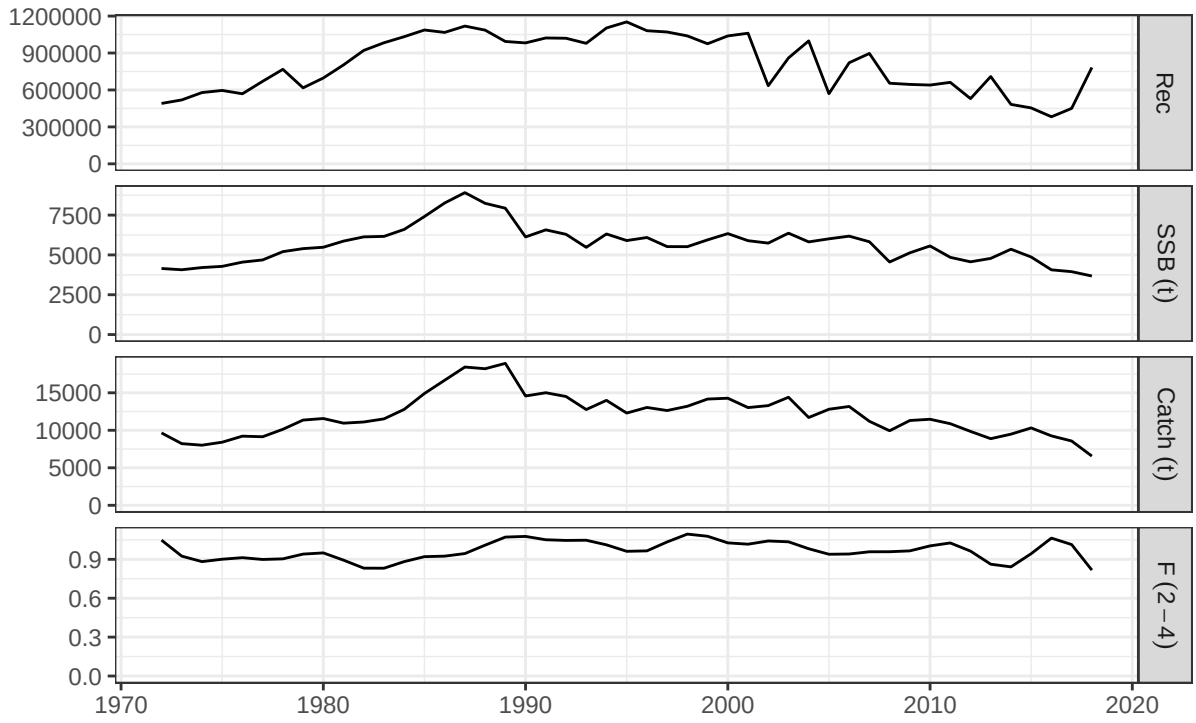


Figure 3. Results for $f: \sim, s(\text{year}, k = 23) + s(\text{age}, k = 4)$.

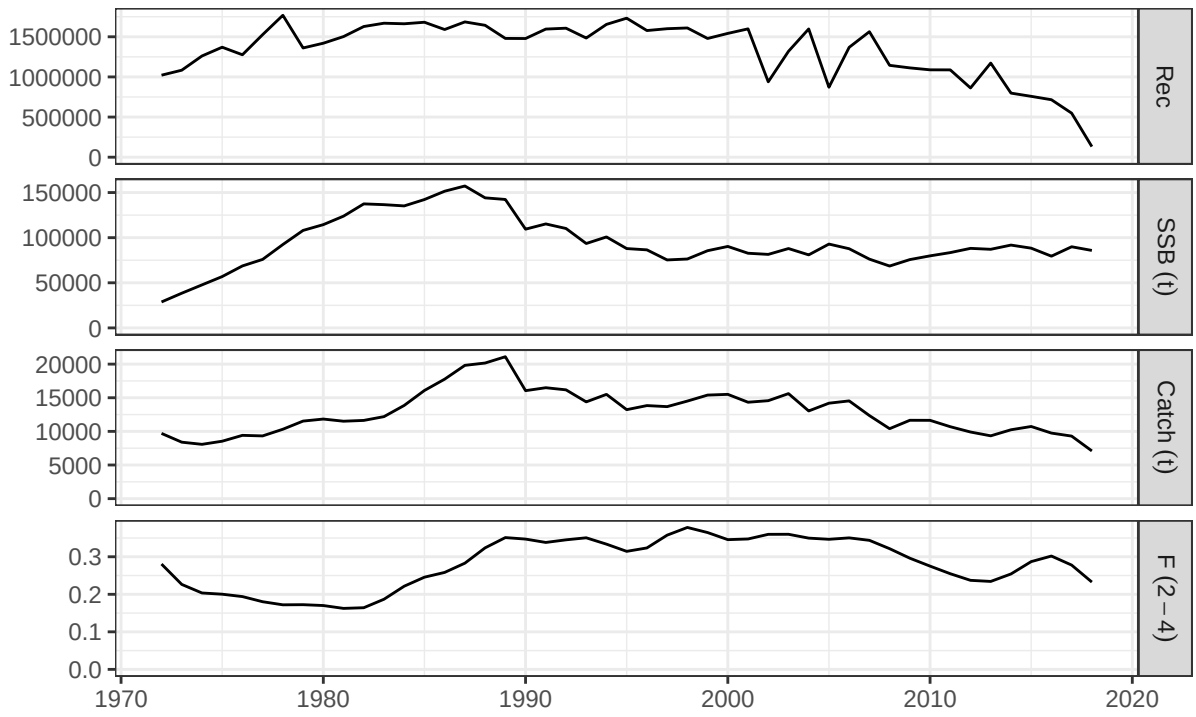


Figure 4. Results for $f: \sim, s(\text{year}, k = 23) + s(\text{age}, k = 3)$.

and the corresponding fishing mortalities:

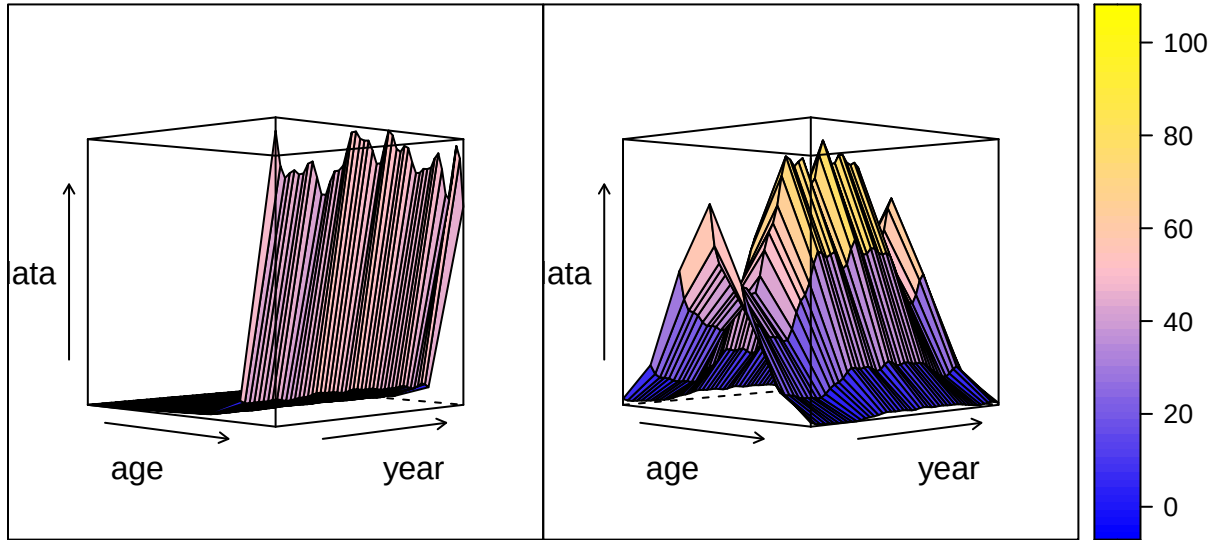


Figure 5. Harvest for fit1 (left) and fit2(right).

3.2 Diagnostics

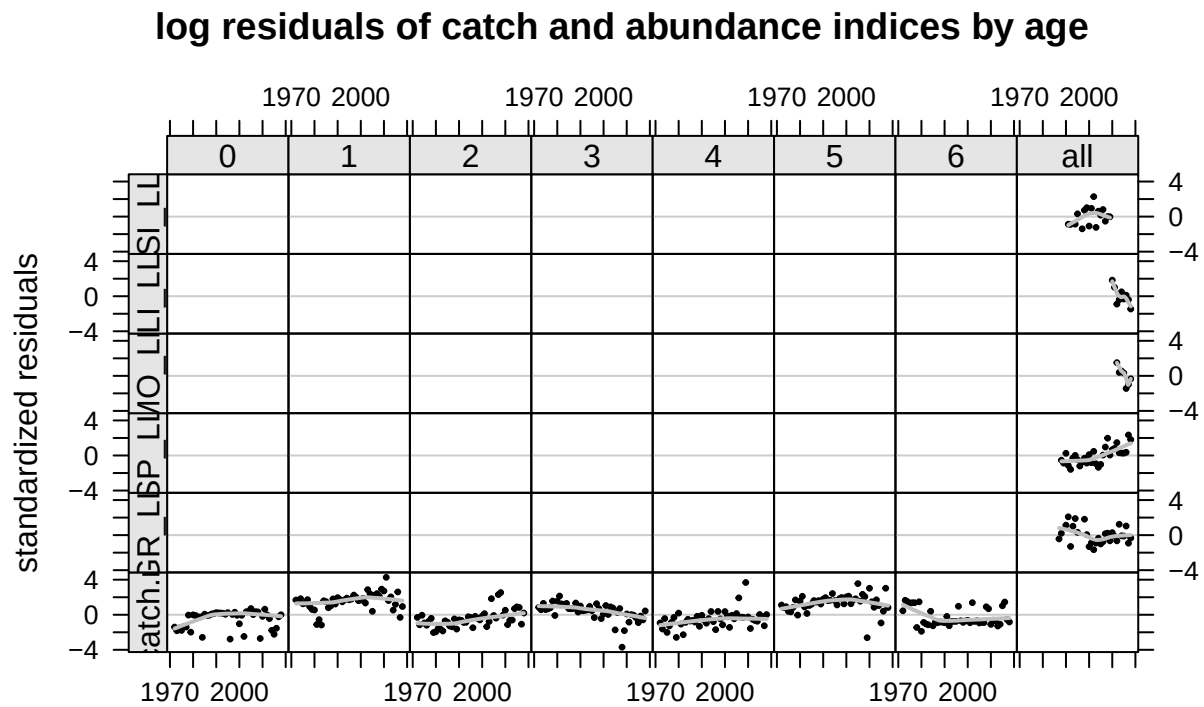


Figure 6. Residuals for fit 1.

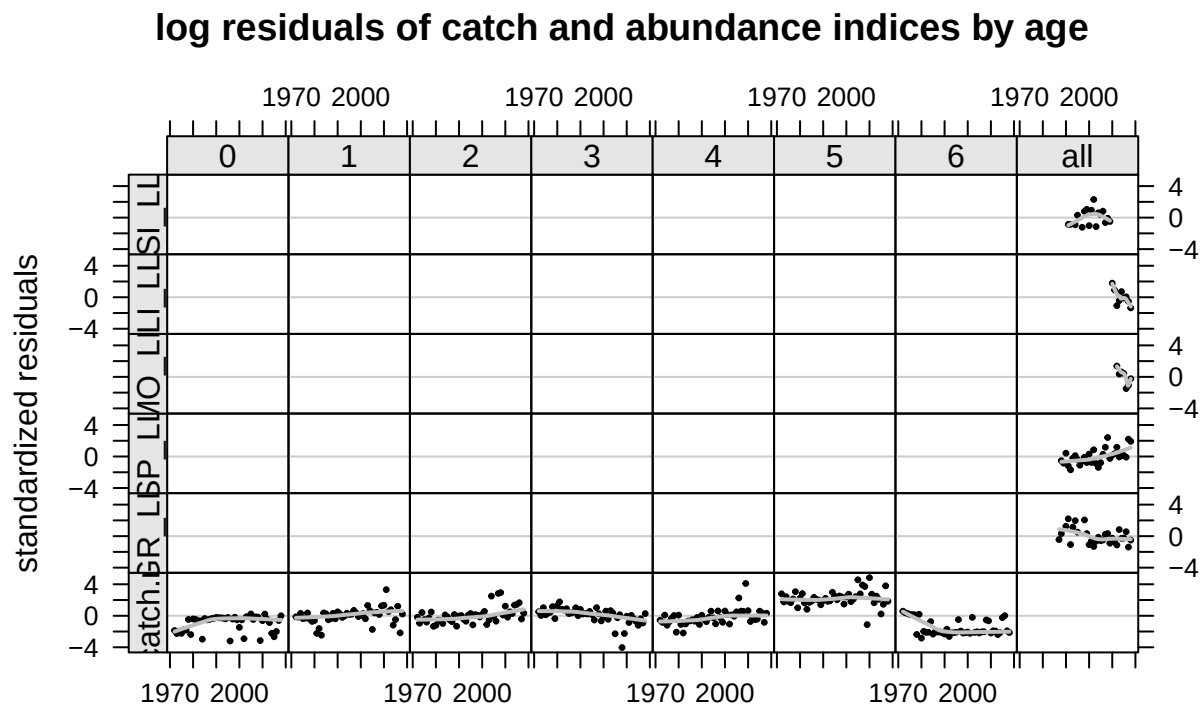


Figure 7. Residuals for fit 2.

3.3 Forcing f to be the same after age 4

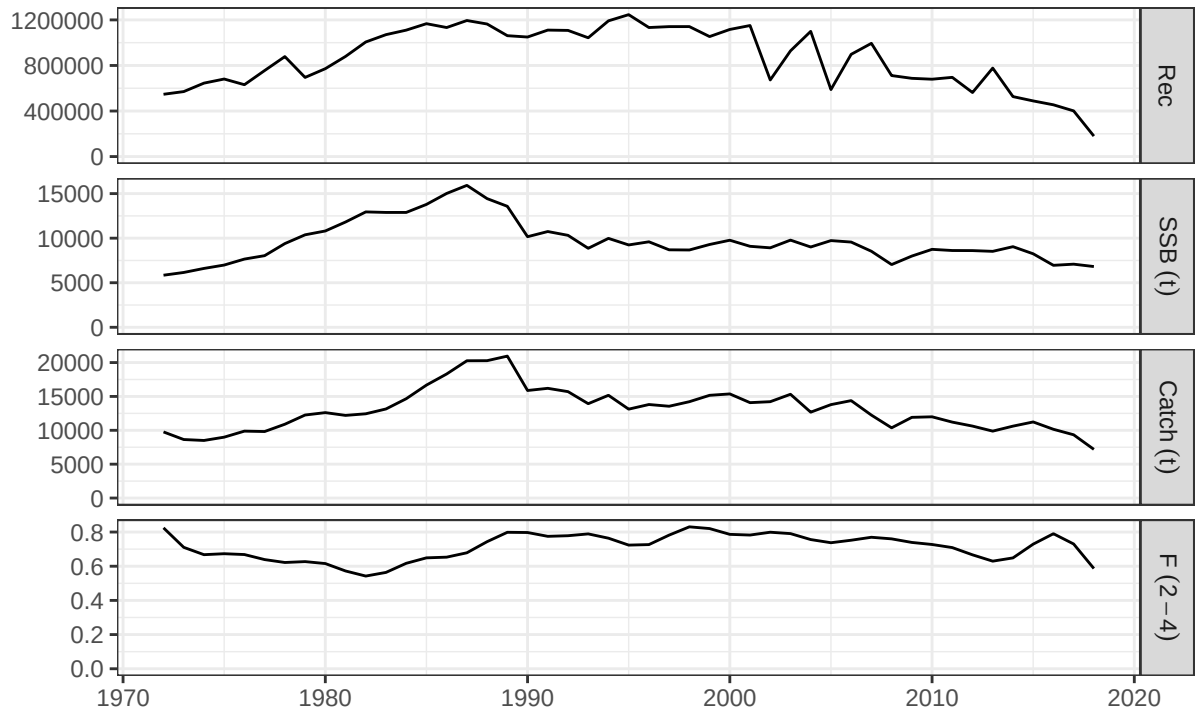


Figure 8. Results for f : $\sim, s(\text{replace}(\text{age}, \text{age} > 4, 4), k = 4) + s(\text{year}, k = 23)$.

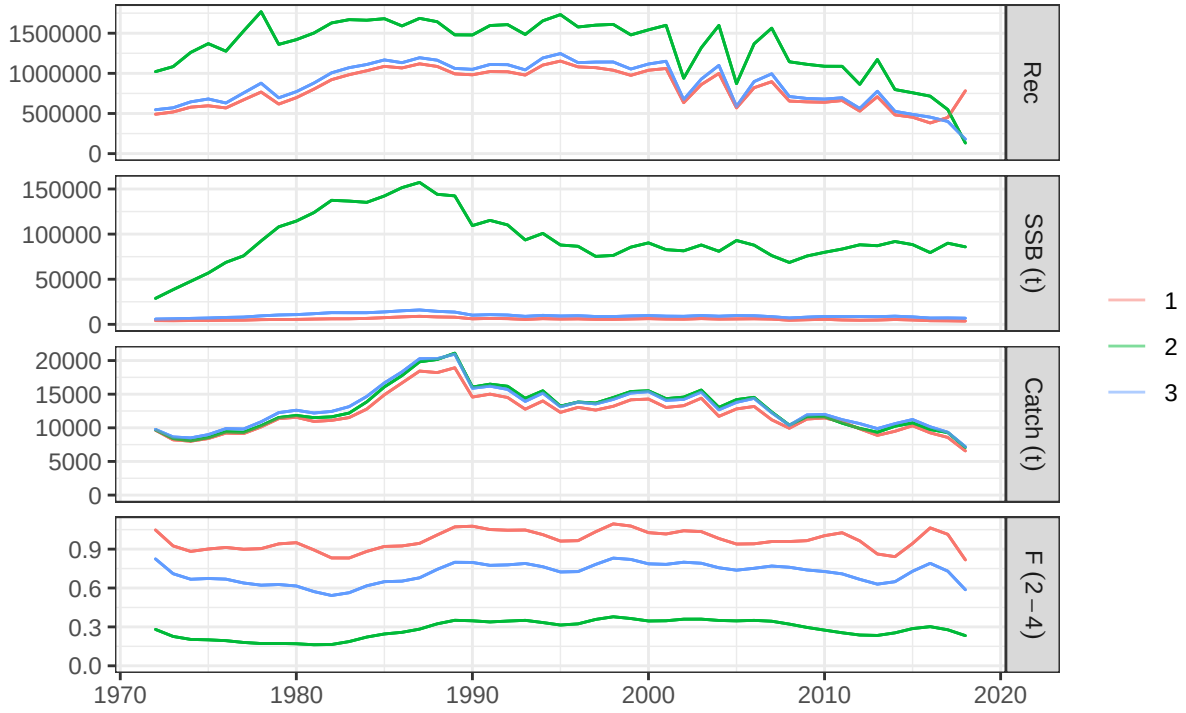


Figure 11. Results for the 3 fits.

Add a smoother to the last run for SR $s(year, k = 20)$:

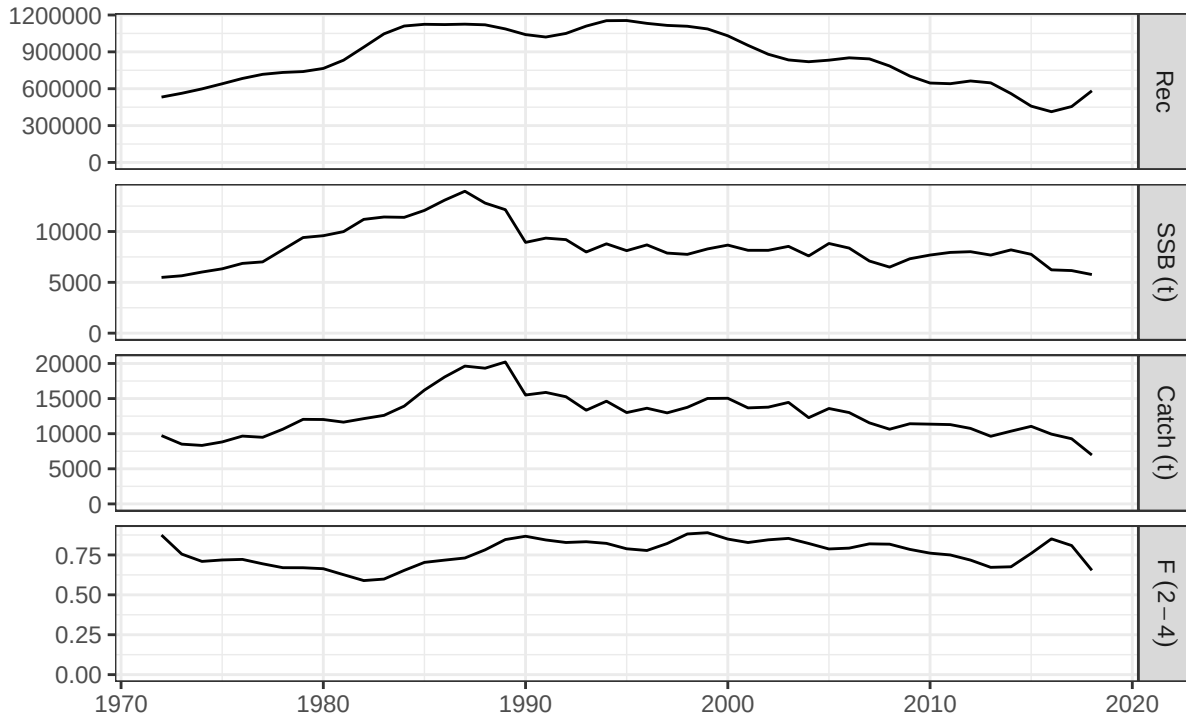


Figure 12. Results for $f: \sim, s(\text{replace}(\text{age}, \text{age} > 4, 4), k = 4) + s(\text{year}, k = 23)$.

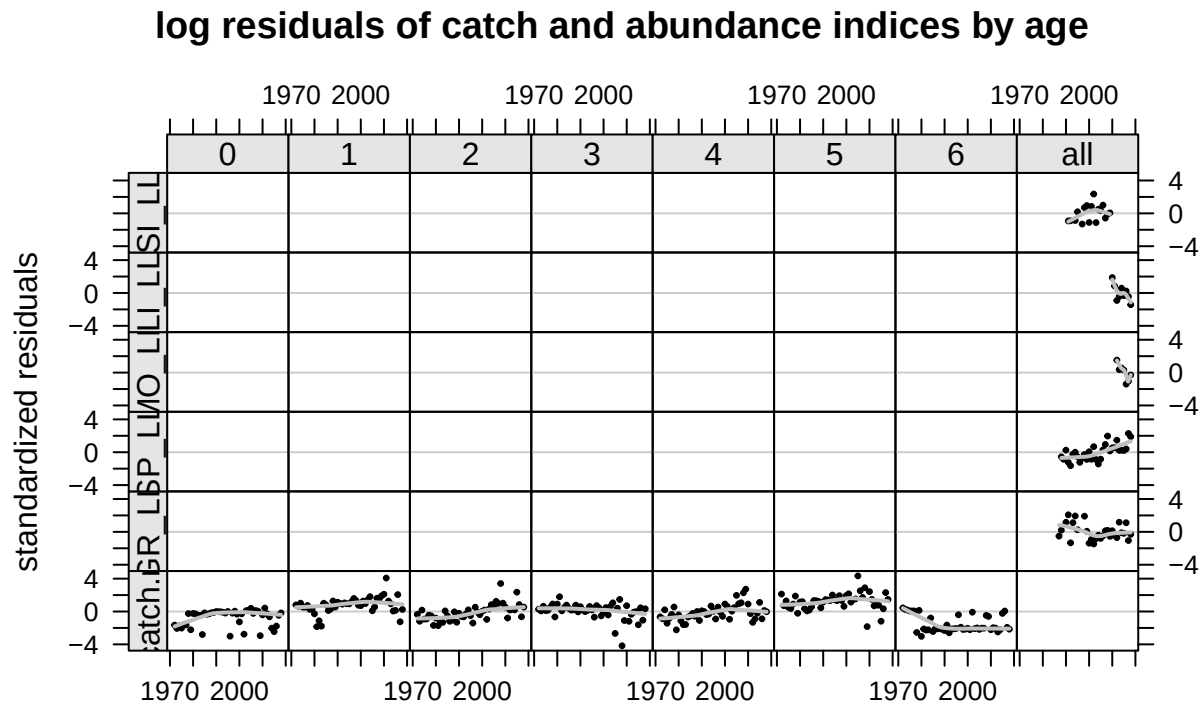


Figure 13. Residuals for fit 4.

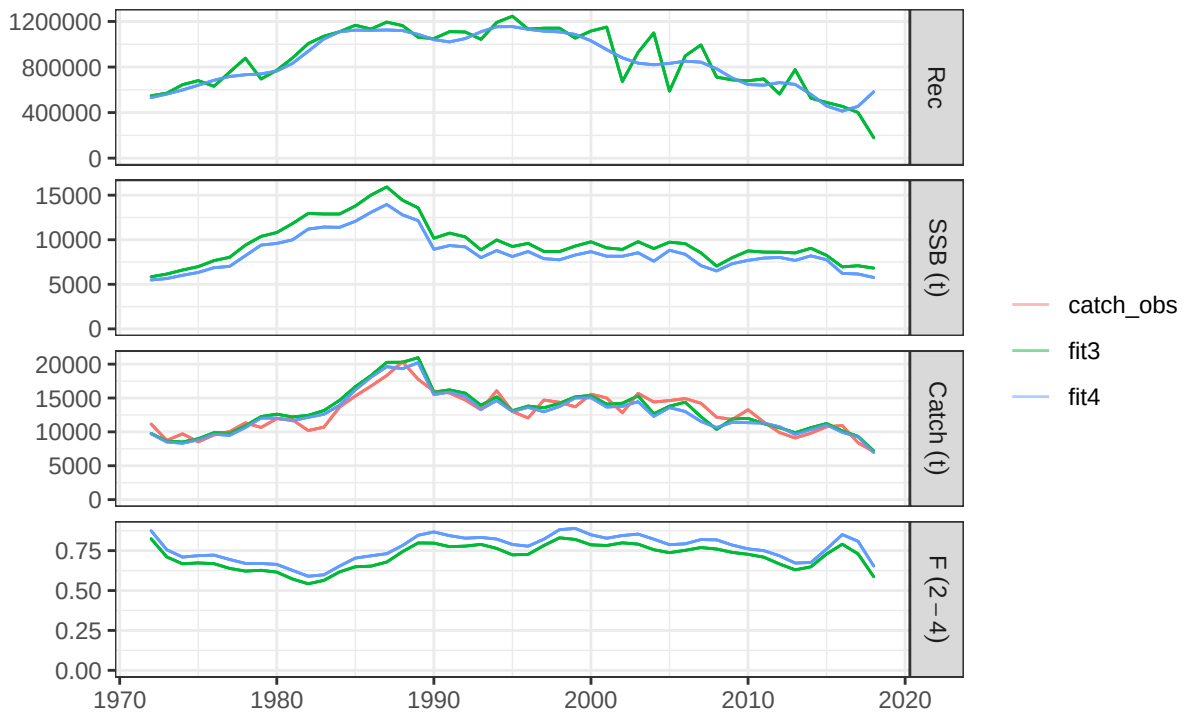


Figure 14. Results for fit3 and fit4.

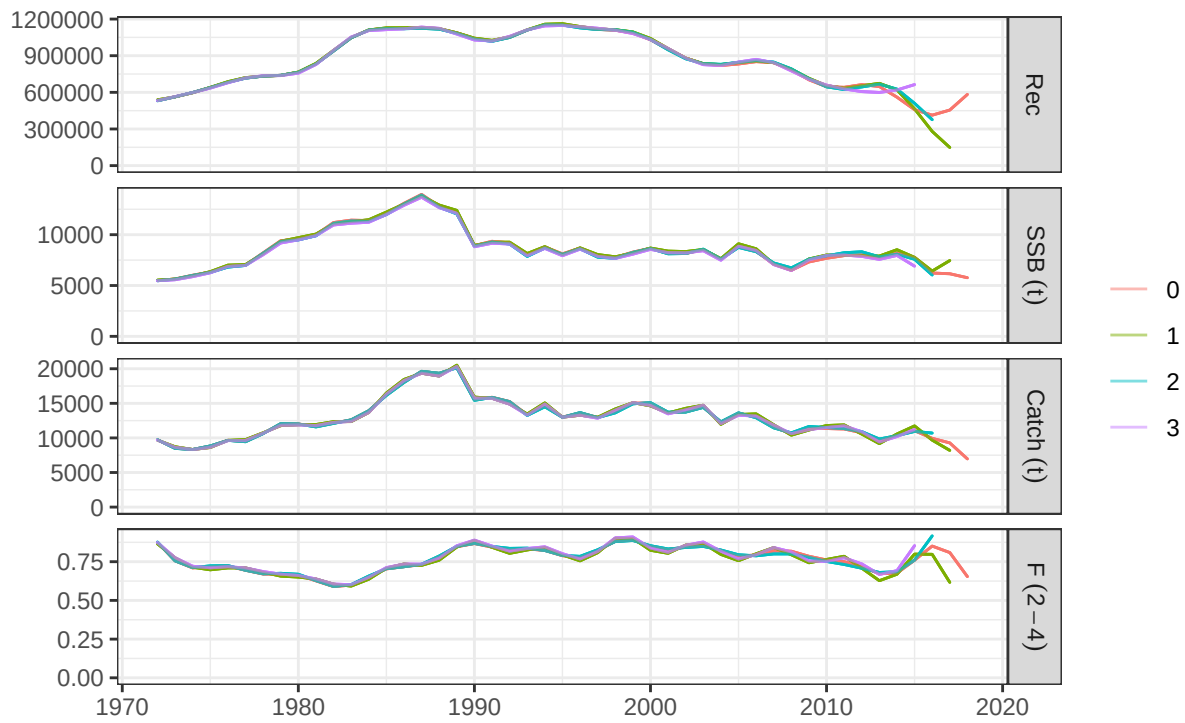


Figure 15. Retrospective plot for fit4.

3.4 Trimmed time series (from 1985)

[...]